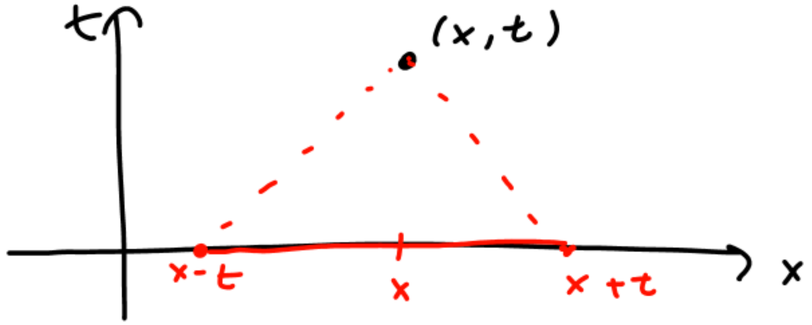
Wave Equation

$n=1 \qquad u_{tt} - u_{xx} = 0$

Last time: Solution formula for $t>0$, $x \in \mathbb{R}$

$$u(x,t) = \frac{1}{2}\left[g(x+t) + g(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy$$

$$\begin{cases} u_{tt} - u_{xx} = 0 \\ u(x,0) = g(x) \\ u_t(x,0) = h(x) \end{cases}$$



Solution on $t>0$, $x>0$ with $u(0,t)=0$

$$u(x,t) = \begin{cases} \frac{1}{2}\left[g(x+t) + g(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy, & x \geq t \\ \frac{1}{2}\left[g(x+t) - g(t-x)\right] + \frac{1}{2}\int_{t-x}^{t+x} h(y)\,dy, & 0 \leq x < t. \end{cases}$$

Solution for $x \in \mathbb{R}^3$, $t > 0$:

$$\begin{cases} u_{tt} - \Delta u = 0 \\ u(x,0) = g(x) \\ u_t(x,0) = h(x) \end{cases}$$

Method of Spherical Means:

Let (x,t) be fixed.

For $r > 0$, define

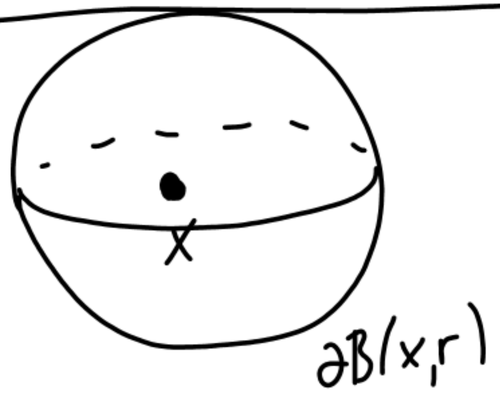
$$U(x,t,r) = U(r) = \fint_{\partial B(x,r)} u(y,t)\, dS(y),$$

$$G(r) = \fint_{\partial B(x,r)} g(y)\, dS, \quad H(r) = \fint_{\partial B(x,r)} h(y)\, dS$$

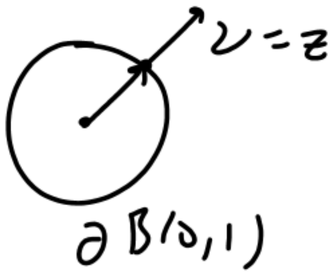
<u>Lemma</u> If u is a C^2 solution to $u_{tt} - \Delta u = 0$, then

$$\begin{cases} U_{tt} - U_{rr} - \frac{2}{r} U_r = 0, \\ U(0,r) = G(r), \quad U_t(0,r) = H(r) \end{cases}$$

$$U(t,r) = \fint_{\partial B(x,r)} u(y,t)\, dS(y)$$



<u>Proof</u>: $$U_{tt} = \frac{\partial^2}{\partial t^2}\left(\fint_{\partial B(x,r)} u(y,t)\,dS_y\right) = \fint_{\partial B(x,r)} u_{tt}\,dS_y$$



$$U_r = \frac{\partial}{\partial r}\left(\fint_{\partial B(x,r)} u(y,t)\,dS_y\right) = \frac{\partial}{\partial r}\left(\frac{1}{4\pi r^2}\int_{\partial B(x,r)} u(y,t)\,dS_y\right)$$

$$= \frac{\partial}{\partial r}\left(\frac{1}{4\pi}\int_{\partial B(0,1)} u(x+rz,t)\,dS_z\right)$$

$y = x + rz$, $\quad z = \frac{y-x}{r}$

$dS_y = r^2\,dS_z$

$$= \frac{1}{4\pi}\int_{\partial B(0,1)} Du(x+rz,t)\cdot z\,dS_z = \frac{r}{4\pi}\int_{B(0,1)} \Delta u(x+rz,t)\,dz$$

$$U_r = \frac{r}{4\pi} \int_{B(0,1)} \Delta u(x+rz, t)\, dz$$

$$y = x + rz$$
$$dy = r^3 dz$$

$$= \frac{r}{4\pi} \int_{B(x,r)} \Delta u(y,t) \frac{dy}{r^3}$$

$$= \frac{r}{3} \fint_{B(x,r)} \Delta u(y,t)\, dy$$

$$= \frac{r}{3} \fint_{B(x,r)} u_{tt}(y,t)\, dy$$

Note: $\lim_{r \to 0} U_r = \lim_{r \to 0} \frac{r}{3} \fint_{B(x,r)} u_{tt}(y,t)\, dy$

$= 0$ since u_{tt} is Continuous.

$$U_r = \frac{r}{3} \fint_{B(x,r)} u_{tt}(y,t)\,dy$$

$$= \frac{1}{4\pi r^2} \int_{B(x,r)} u_{tt}(y,t)\,dy$$

Fact:

$$\frac{\partial}{\partial r}\left(\int_{B(x,r)} f(y)\,dy\right) = \int_{\partial B(x,r)} f(y)\,dS_y$$

if f is a smooth function.

$$U_{rr} = \frac{\partial}{\partial r}\left(\frac{1}{4\pi r^2} \int_{B(x,r)} u_{tt}(y,t)\,dy\right)$$

$$= \frac{-2}{4\pi r^3} \int_{B(x,r)} u_{tt}(y,t)\,dy + \frac{1}{4\pi r^2} \int_{\partial B(x,r)} u_{tt}(y,t)\,dS_y = \frac{-2}{3} \fint_{B(x,r)} u_{tt}(y,t)\,dy$$

$$+ \fint_{\partial B(x,r)} u_{tt}(y,t)\,dS_y$$

$$U_{tt} - U_{rr} = \fint_{\partial B(x,r)} u_{tt}\, dS_y + \frac{2}{3} \fint_{B(x,r)} u_{tt}\, dy - \fint_{\partial B(x,r)} u_{tt}\, dS_y$$

$$= \frac{2}{3} \fint_{B(x,r)} u_{tt}\, dy = \frac{2}{r} U_r \qquad \left(\text{since } U_r = \frac{r}{3} \fint_{B(x,r)} u_{tt}\, dy \right)$$

$$\rightarrow \quad U_{tt} - U_{rr} - \frac{2}{r} U_r = 0 .$$

Note: $\lim_{r \to 0} U(x,t,r) = \lim_{r \to 0} \fint_{\partial B(x,r)} u(y,t)\, dS_y = u(x,t)$.

Next, define $\tilde{U} = rU$

$$\tilde{U}_{tt} = (rU)_{tt} = rU_{tt}$$

$$\tilde{U}_r = (rU)_r = U + rU_r$$

$$\tilde{U}_{rr} = (U + rU_r)_r = 2U_r + rU_{rr}$$

$$\tilde{U}_{tt} - \tilde{U}_{rr} = rU_{tt} - 2U_r - rU_{rr}$$

$$= r\left(U_{tt} - \frac{2}{r}U_r - U_{rr}\right) = 0$$

$\tilde{U}$ solves 1D wave equation.

As $r \to 0$,

$$\tilde{U}(x,t,r) = \underbrace{r}_{0}\,\underbrace{U(x,t,r)}_{u(x,t)}$$

$$\tilde{U} \longrightarrow 0$$

$\to$ $\tilde{U}$ satisfies the right BC at $r=0$ to do odd reflection.

$$\tilde{U}(x,t,r) = r\,U(x,t,r)$$

satisfies

$$\tilde{U}(x,0,r) = r\,G(r) = \tilde{G}(r)$$

$$\tilde{U}_t(x,0,r) = r\,H(r) = \tilde{H}(r)$$

$\rightarrow$ Solution formula on half-line:

$$\tilde{U}(x,t,r) = \frac{1}{2}\left[\tilde{G}(r+t) - \tilde{G}(t-r)\right] + \frac{1}{2}\int_{t-r}^{t+r} \tilde{H}(y)\,dy, \quad 0 \le r < t$$

Divide by r:

$$U(x,t,r) = \frac{\tilde{G}(r+t) - \tilde{G}(t-r)}{2r} + \frac{1}{2r}\int_{t-r}^{t+r} \tilde{H}(y)\,dy$$

Take limit as $r \to 0$:

$$u(x,t) = \lim_{r\to 0} \frac{\tilde{G}(t+r) - \tilde{G}(t-r)}{2r} + \lim_{r\to 0} \fint_{t-r}^{t+r} \tilde{H}(y)\,dy$$

For any diff'ble function $f(x)$,

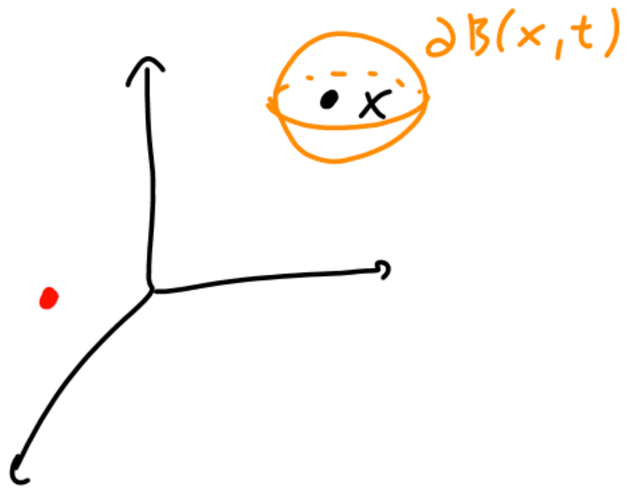
$$\lim_{r\to 0} \frac{f(x+r)-f(x-r)}{2r} = \frac{1}{2}\lim_{r\to 0}\left(\frac{f(x+r)-f(x)}{r} + \frac{f(x)-f(x-r)}{r}\right)$$

$$= \frac{1}{2}\, 2 f'(x) = f'(x)$$

$$u(x,t) = \widehat{G}'(t) + \widehat{H}(t)$$

$$= \frac{\partial}{\partial t}\Big(t\, G(t)\Big) + t\, H(t)$$

$$= \frac{\partial}{\partial t}\left(t \fint_{\partial B(x,t)} g(y)\, dS_y\right) + t \fint_{\partial B(x,t)} h(y)\, dS_y$$



$$\frac{\partial}{\partial t}\left(t \fint_{\partial B(x,t)} g(y)\,dS_y\right) = \frac{\partial}{\partial t}\left(\frac{t}{4\pi t^2}\int_{\partial B(x,t)} g(y)\,dS_y\right)$$

$$x + tz = y$$

$$tz = y - x$$

$$dS_y = t^2\,dS_z$$

$$= \frac{\partial}{\partial t}\left(\frac{t}{4\pi}\int_{\partial B(0,1)} g(x+tz)\,dS_z\right)$$

$$= \frac{1}{4\pi}\int_{\partial B(0,1)} g(x+tz)\,dS_z + \frac{t}{4\pi}\int Dg(x+tz)\cdot z\,dS_z$$

$$= \fint_{\partial B(x,t)} g(y)\,dS_y + \fint_{\partial B(x,t)} Dg(y)\cdot(y-x)\,dS_y$$

$$\boxed{u(x,t) = \fint_{\partial B(x,t)} \left[g(y) + Dg(y)\cdot(y-x) + t\,h(y)\right] dS_y}$$

Kirchoff's Formula

3D Wave Equation

Method of Spherical Means doesn't work in $n=2$.

Instead, use Method of Descent.

Idea: Given a solution to 2D Wave Equation

$$u_{tt} - \Delta u = 0$$

$$u_{tt} - u_{xx} - u_{yy} = 0,$$

Extend to 3 space dimension

$$\bar{u}(t,x,y,z) = u(t,x,y)$$

Since $\bar{u}_{zz} = 0$, we have

$$\bar{u}_{tt} - \bar{u}_{xx} - \bar{u}_{yy} - \bar{u}_{zz}$$

$$= u_{tt} - u_{xx} - u_{yy} - 0 = 0$$

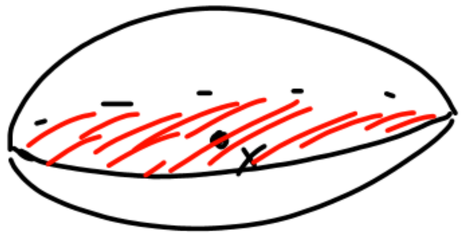
$\rightarrow$ $\bar{u}$ solves the 3D Wave Equation.

Extend g, h similarly:

$$\bar{g}(x_1, x_2, x_3) = g(x_1, x_2)$$

$$\bar{h}(x_1, x_2, x_3) = h(x_1, x_2)$$

Kirchoff:

$$\bar{u}(\vec{x}, t) = \frac{\partial}{\partial t}\left(t \fint_{\partial B(x,t)} \bar{g}(y)\, dS_y\right) + t \fint_{\partial B(x,t)} \bar{h}(y)\, dS_y$$

$$\fint_{\partial B(x_1, x_2, x_3, t)} \bar{g}(y)\, dS_y = \fint_{\partial B(x,t)} g(y_1, y_2)\, dS_y$$